

PRD_Assets_Module

Enterprise Product Requirements Document

Audience: Full Stack Developers, Product, UX/UI, QA, DevOps

Purpose: Define the complete architecture, workflows, UI, backend behavior, integrations and implementation requirements for the Assets module.

1. Product Vision

Build an enterprise-grade asset lifecycle management system integrated with Finance, Procurement, Projects, Documents, HR, Contracts and Platform Configuration. Every asset is traceable from acquisition to disposal.

2. Information Architecture

Assets

- Dashboard
- Assets Module
- Asset Page
- Categories
- Hierarchy
- Ownership
- Assignment
- Locations
- QR / Barcode / RFID
- Maintenance
- Work Orders
- Inspections
- Calibration
- Warranty
- Insurance
- Reservations
- Transfers
- Documents
- Activity Timeline
- Reports
- Settings

3. Assets Module

Features:

- Global search
- Advanced filters
- Saved views

- List/Card/Table layouts
- Import/Export
- Bulk actions
- KPI cards
- Personal favorites
- Responsive desktop/tablet/mobile

4. Asset Page

Sections:

- Overview
- Technical specifications
- Financial summary
- Ownership
- Assignment history
- Current location
- QR/Barcode/RFID
- Maintenance
- Work orders
- Inspections
- Warranty
- Insurance
- Documents
- Linked Projects
- Linked Contracts
- Linked Tasks
- Activity timeline

5. Asset Lifecycle

Draft → Active → Assigned → Reserved → Maintenance → Out of Service → Retired → Disposed → Archived

6. Maintenance

Preventive, corrective and emergency maintenance.

Recurring schedules.

Automatic work order generation.

Technician assignment.

Labor, materials, spare parts, downtime, costs and approvals.

7. Work Orders

Lifecycle:

Draft → Approved → Assigned → In Progress → Waiting Parts → Waiting Approval → Completed → Closed → Archived

Fields:

Number, Asset, Priority, SLA, Dates, Labor, Materials, Photos, Attachments, Checklist, Signature, Timeline.

8. Inspections & Calibration

Template-based inspections, pass/fail, photos, certificates, calibration schedules, tolerance limits and compliance history.

9. Warranty & Insurance

Coverage, provider, claims, expiry alerts, linked repairs, policy tracking and renewal reminders.

10. Asset Transfers

Transfer between companies, departments, teams, users, projects and warehouses with complete audit history.

11. Finance Integration

Acquisition cost, depreciation, book value, maintenance cost, disposal value, project cost allocation and links to Finance journals.

12. Procurement Integration

Purchase requisitions, purchase orders, suppliers, receiving, invoices and procurement history.

13. Project Integration

Assign assets to projects, track utilization, cost allocation, reservations and availability.

14. Documents Integration

Specifications, manuals, invoices, certificates, warranties, inspection reports, photos and revisions.

15. Compliance & Risk

Safety compliance, regulatory inspections, certifications, risks, incidents and mitigation.

16. Reports

Asset register, depreciation, maintenance cost, downtime, utilization, warranties, insurance, inspections, compliance and executive dashboards.

17. Responsive UX

Desktop workspace, adaptive tablet layout and mobile-first operational workflows.

18. Premium UX

Hover elevation, smooth transitions, skeleton loading, animated timelines, subtle KPI transitions and reduced-motion support.

19. Platform Configuration

Every module, page, widget, workflow, report and action must support enable/disable through Platform Configuration.

20. Database Entities

Asset
AssetCategory
AssetAssignment
AssetLocation
AssetMaintenance
WorkOrder
Inspection
Calibration
Warranty
InsurancePolicy
Reservation
Transfer
AssetDocument
AssetActivity
AssetReport

21. REST API

GET /assets
POST /assets
GET /assets/{id}
PATCH /assets/{id}
POST /assets/{id}/assign
POST /assets/{id}/transfer
POST /assets/{id}/maintenance
POST /work-orders
GET /reports/assets

22. Validation Rules

Unique asset code, mandatory ownership, valid lifecycle transitions, maintenance dependencies, transfer validation, immutable audit history.

23. Security

Tenant isolation, company scope, audit logging, permission-ready architecture and confidential financial data separation.

24. Testing

Unit, integration, end-to-end, performance, accessibility and responsive testing.

25. Acceptance Criteria

- Complete asset lifecycle
- Enterprise maintenance engine
- Finance integration
- Procurement integration
- Document integration
- Audit trail
- Responsive UI
- Premium interactions
- Platform Configuration compatibility
- Scalable architecture